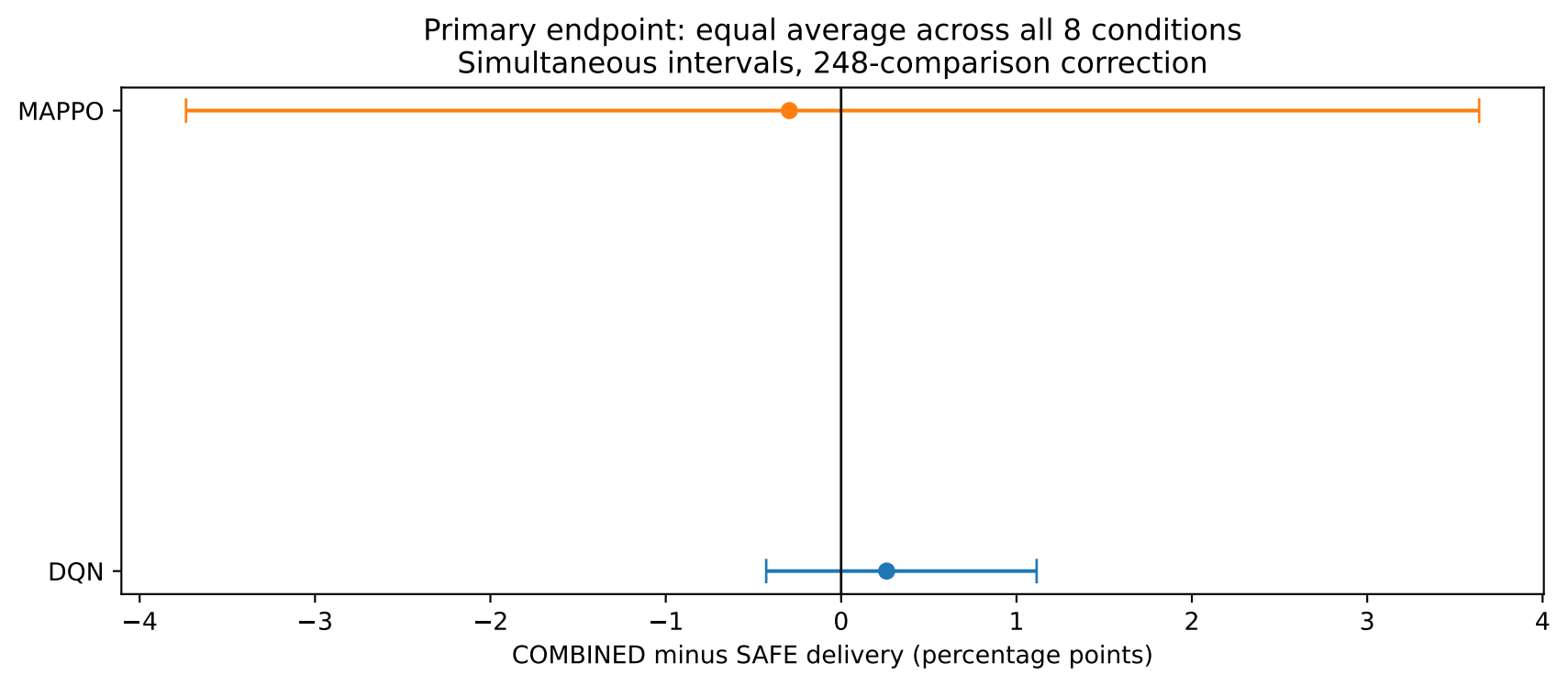
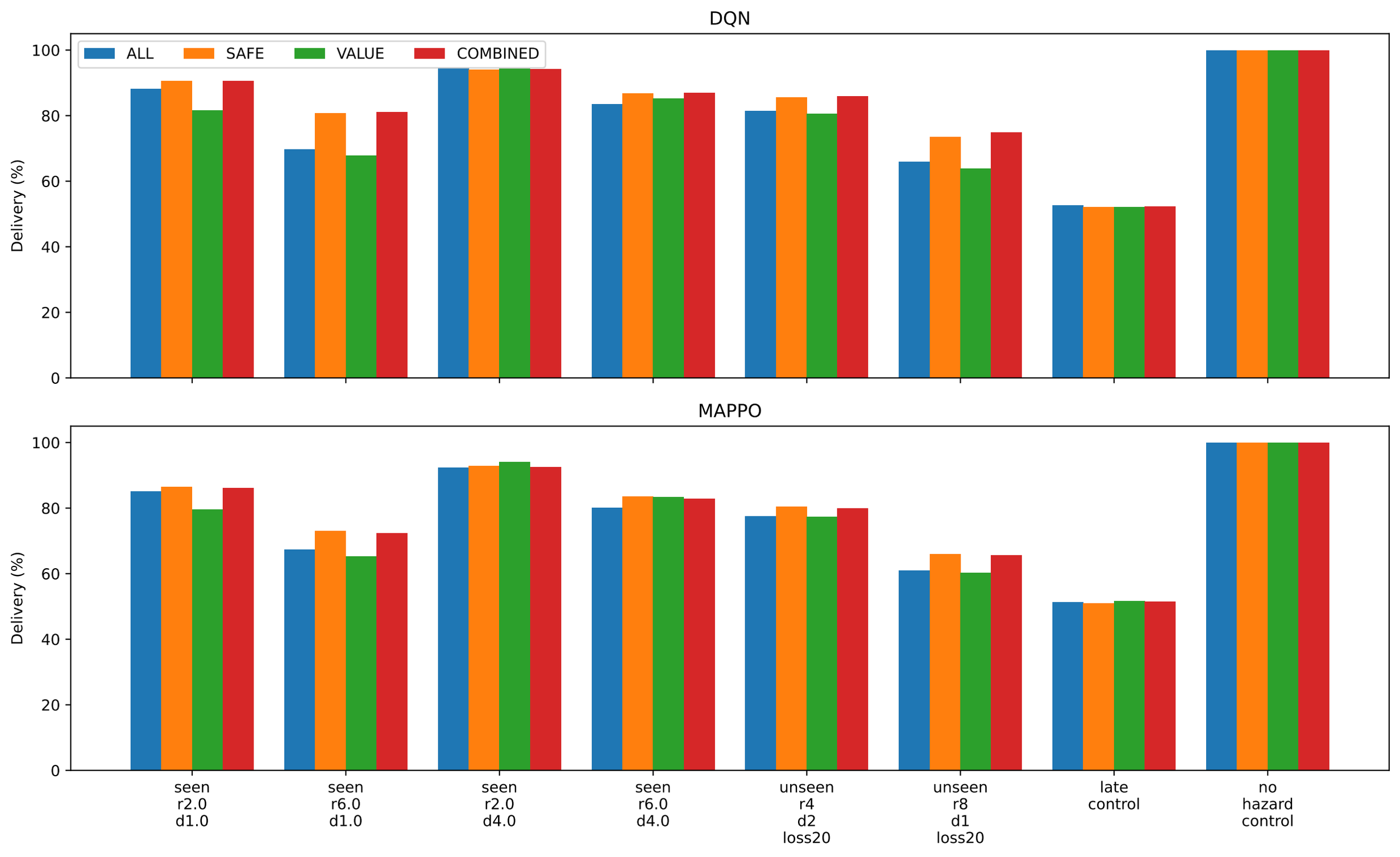
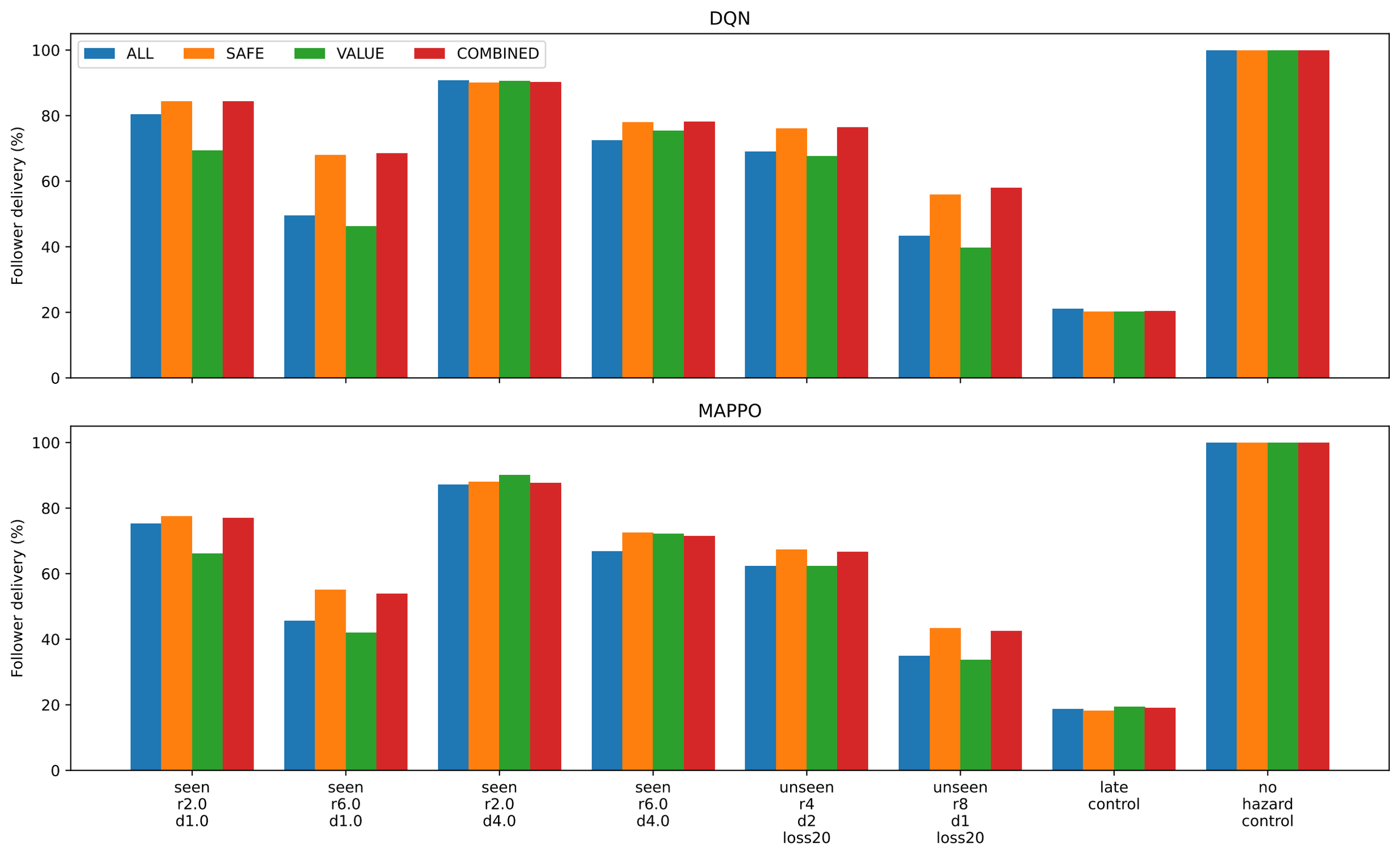
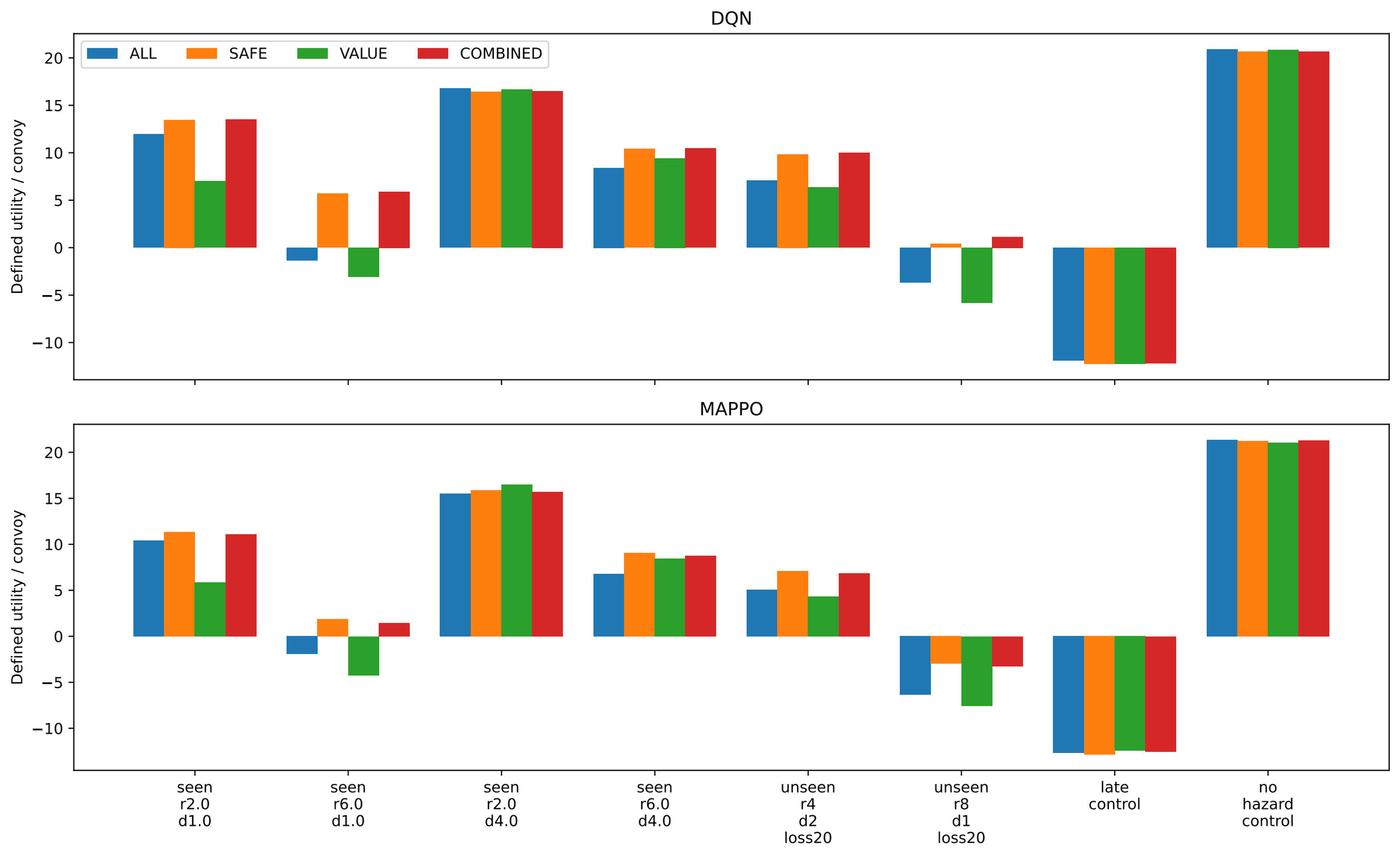
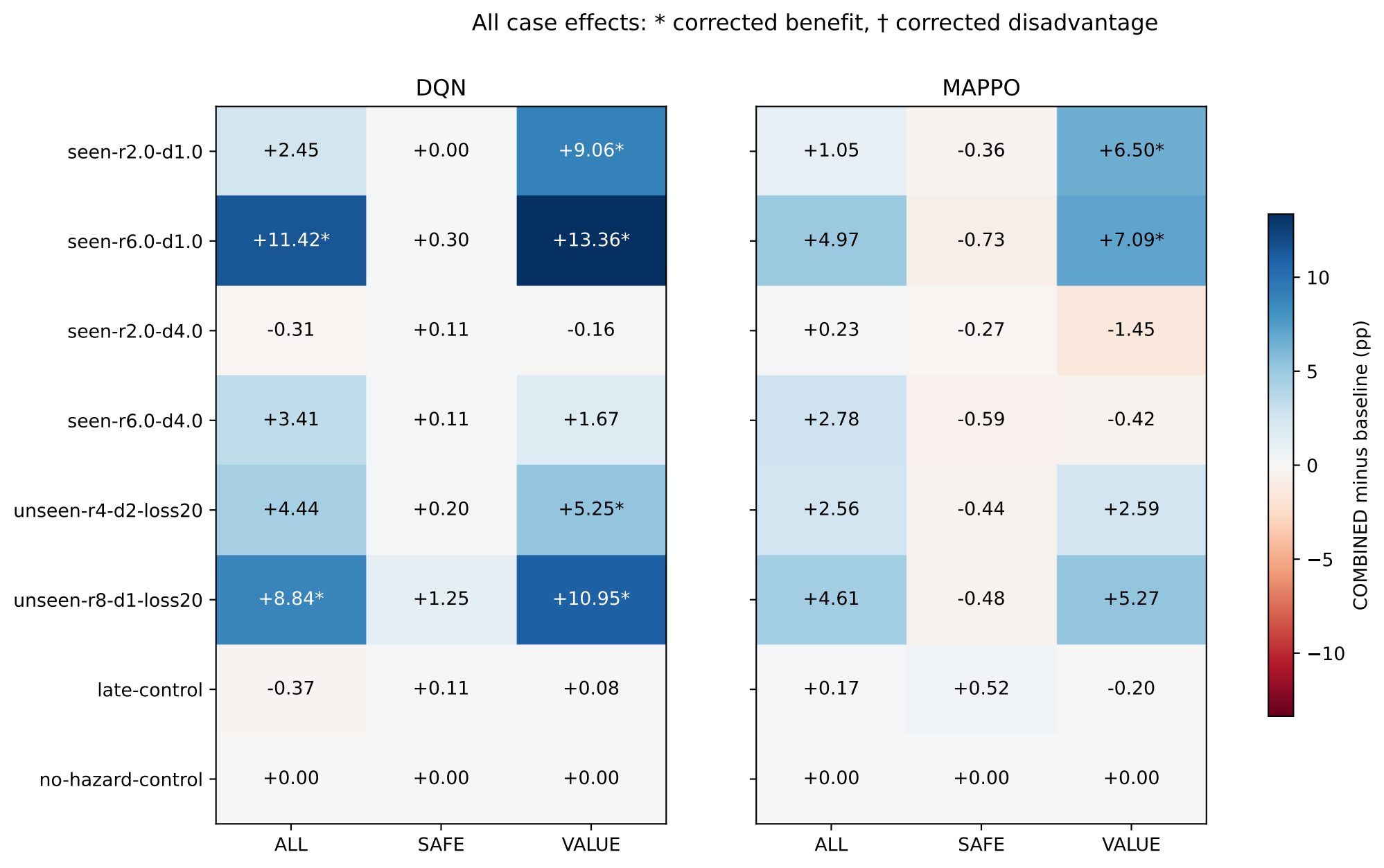
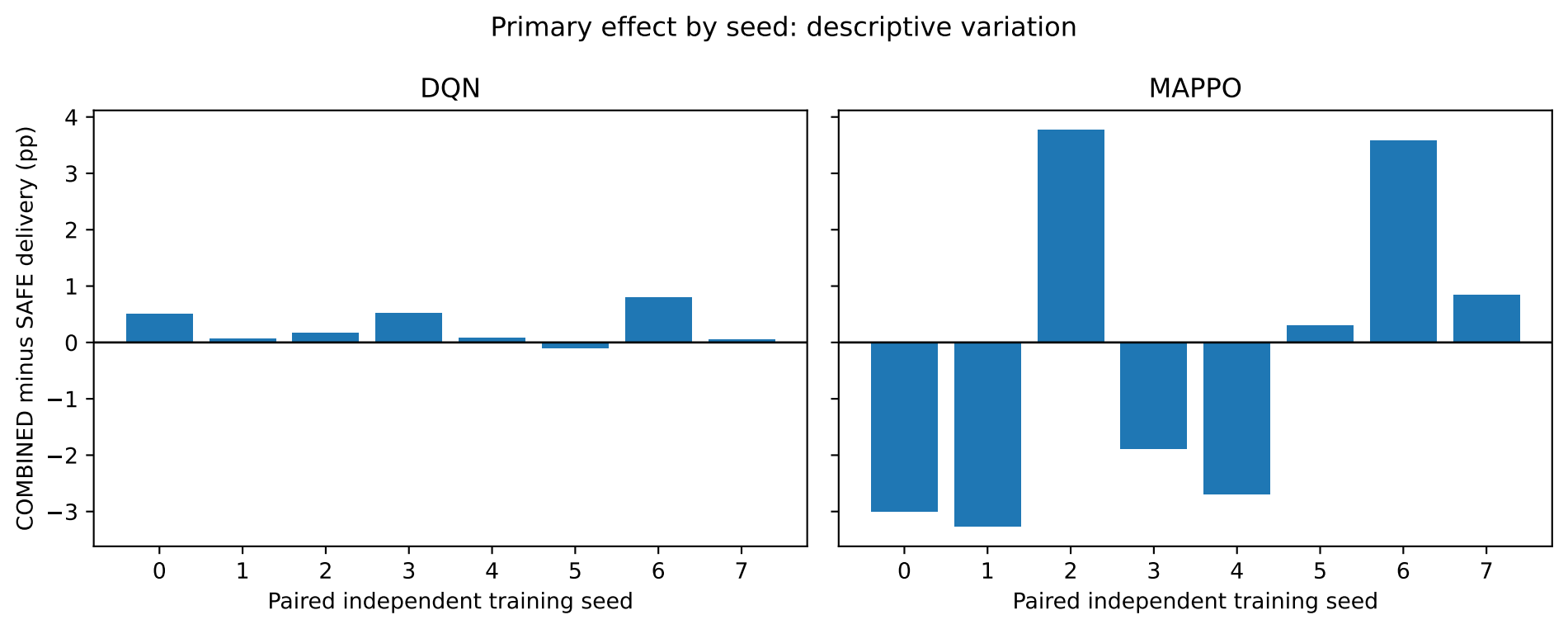

Primary endpoint: equal average across all 8 conditions
Simultaneous intervals, 248-comparison correction
MAPPO
DQN
−4
−3
−2
−1
0
1
2
3
4
COMBINED minus SAFE delivery (percentage points)

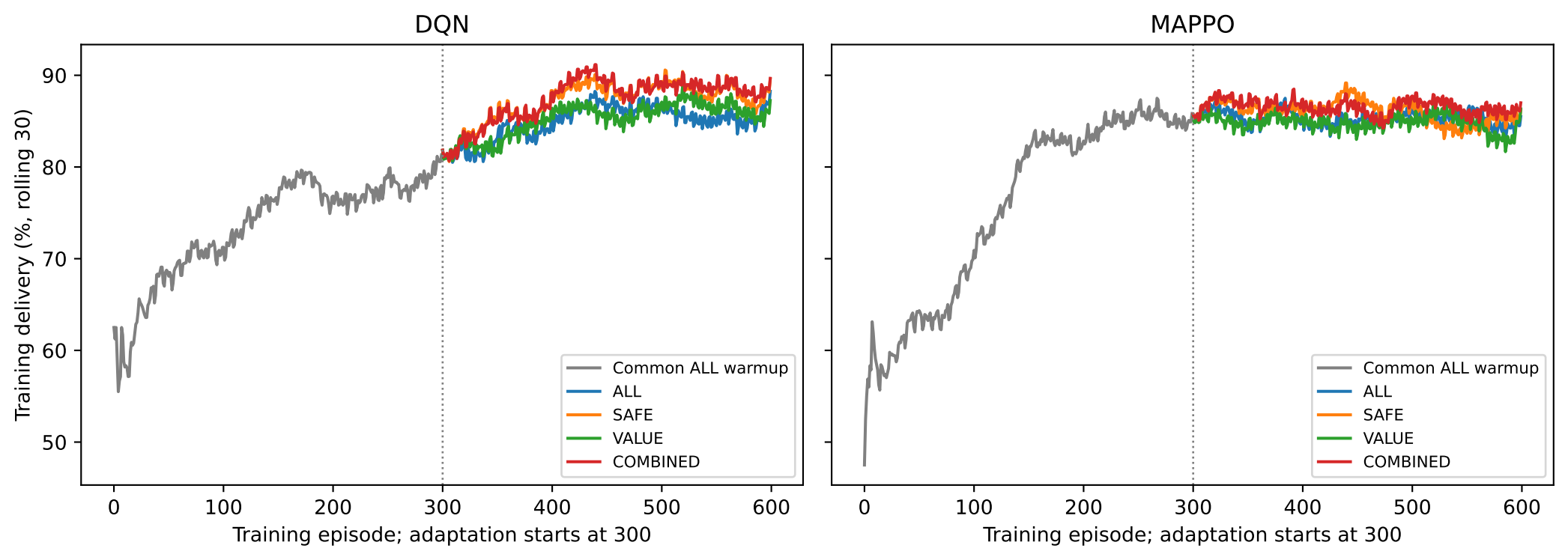

DQN
ALL
SAFE
VALUE
COMBINED
Delivery (%)
0
20
40
60
80
100
MAPPO
Delivery (%)
0
20
40
60
80
100
seen r2.0 d1.0
seen r6.0 d1.0
seen r2.0 d4.0
seen r6.0 d4.0
unseen r4 d2 loss20
unseen r8 d1 loss20
late control
no hazard control

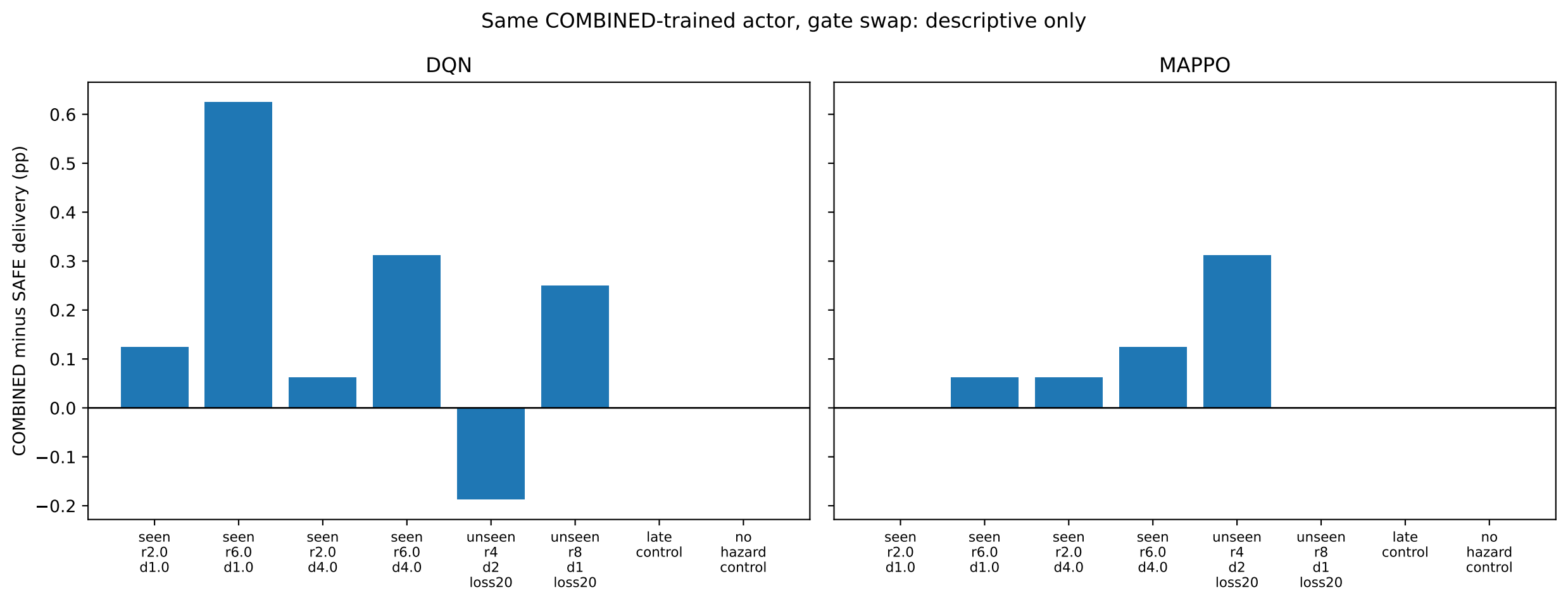

DQN
ALL
SAFE
VALUE
COMBINED
Follower delivery (%)
0
20
40
60
80
100
MAPPO
Follower delivery (%)
0
20
40
60
80
100
seen r2.0 d1.0
seen r6.0 d1.0
seen r2.0 d4.0
seen r6.0 d4.0
unseen r4 d2 loss20
unseen r8 d1 loss20
late control
no hazard control


DQN
ALL
SAFE
VALUE
COMBINED
Defined utility / convoy
20
15
10
5
0
−5
−10
MAPPO
Defined utility / convoy
20
15
10
5
0
−5
−10
seen r2.0 d1.0
seen r6.0 d1.0
seen r2.0 d4.0
seen r6.0 d4.0
unseen r4 d2 loss20
unseen r8 d1 loss20
late control
no hazard control


All case effects: * corrected benefit, † corrected disadvantage
DQN
MAPPO
seen-r2.0-d1.0
seen-r6.0-d1.0
seen-r2.0-d4.0
seen-r6.0-d4.0
unseen-r4-d2-loss20
unseen-r8-d1-loss20
late-control
no-hazard-control
+2.45
+0.00
+9.06*
+11.42*
+0.30
+13.36*
-0.31
+0.11
-0.16
+3.41
+0.11
+1.67
+4.44
+0.20
+5.25*
+8.84*
+1.25
+10.95*
-0.37
+0.11
+0.08
+0.00
+0.00
+0.00
+1.05
-0.36
+6.50*
+4.97
-0.73
+7.09*
+0.23
-0.27
-1.45
+2.78
-0.59
-0.42
+2.56
-0.44
+2.59
+4.61
-0.48
+5.27
+0.17
+0.52
-0.20
+0.00
+0.00
+0.00
ALL
SAFE
VALUE
ALL
SAFE
VALUE
COMBINED minus baseline (pp)
10
5
0
−5
−10


Primary effect by seed: descriptive variation
DQN
MAPPO
COMBINED minus SAFE delivery (pp)
4
3
2
1
0
−1
−2
−3
0
1
2
3
4
5
6
7
Paired independent training seed
Paired independent training seed


DQN
MAPPO
Training delivery (%, rolling 30)
90
80
70
60
50
0
100
200
300
400
500
600
Training episode; adaptation starts at 300
Common ALL warmup
ALL
SAFE
VALUE
COMBINED


Same COMBINED-trained actor, gate swap: descriptive only
DQN
MAPPO
COMBINED minus SAFE delivery (pp)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
seen r2.0 d1.0
seen r6.0 d1.0
seen r2.0 d4.0
seen r6.0 d4.0
unseen r4 d2 loss20
unseen r8 d1 loss20
late control
no hazard control